

Entropy Stable Galerkin Projection Reduced Order Model

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Introduction

- Computational Models have been increasingly essential in design and engineering
- However, high costs and storage requirements lead to bottlenecks

→ Goal: Reduce the size of the problem

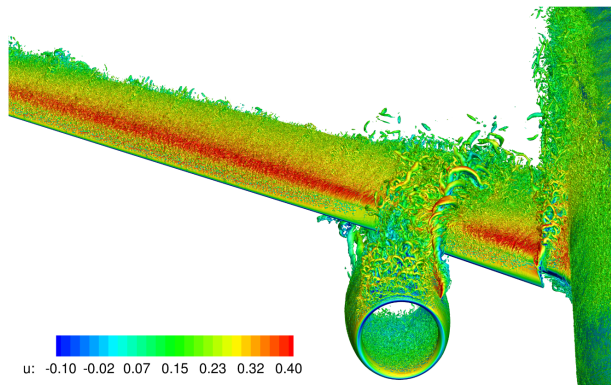


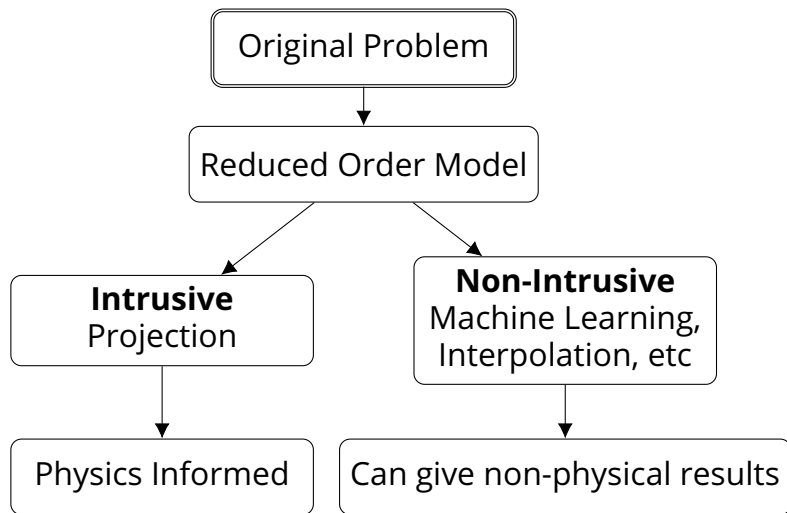
Figure: NASA high-lift common research model at $\alpha = 21.47^\circ$, $M = 0.2$, $Re = 5.49 \times 10^6$, simulated using WMLES.¹

¹Wang, Li et al. *AIAA SciTech 2022 Forum*. 2022.

Reduced Order Modeling

To reduce the size of the problem, we turn to reduced order modeling.

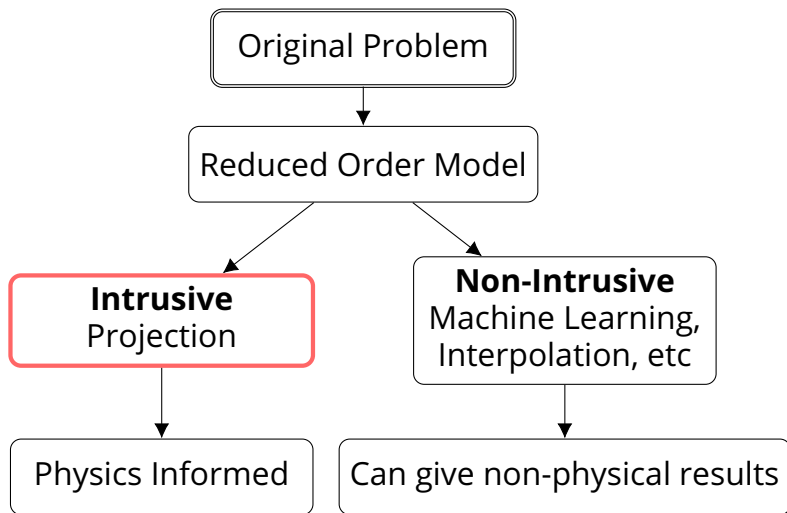
- Reduces the size of the problem
- Solving same equation



Reduced Order Modeling

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- Reduces the size of the problem
- Solving same equation



Projection Based Reduced Order Modeling

We assume that the solution belongs on a lower-dimension subspace, $\mathbf{V} \in \mathbb{R}^{n \times m}$, a trial basis

$$\mathbf{u}_h = \mathbf{u}_{ref} + \mathbf{V}\tilde{\mathbf{u}}$$

We can then project our original equation into the reduced order space by pre-multiplying by the test basis, $\mathbf{W} \in \mathbb{R}^{n \times m}$

$$\mathbf{M} \frac{d\mathbf{u}_h}{dt} = \mathbf{f}(\mathbf{u}^n, t) \longrightarrow \mathbf{W}^T \mathbf{M} \frac{d\mathbf{V}\tilde{\mathbf{u}}}{dt} = \mathbf{W}^T \mathbf{f}(\mathbf{u}_{ref} + \mathbf{V}\tilde{\mathbf{u}}^n, t)$$

$$n \times n \longrightarrow m \times m, m \ll n$$

Entropy Stable Discontinuous Galerkin

We now define a skew-symmetric formulation for the DG Method² in 1D.

$$\mathbf{M} \frac{d\mathbf{u}_h}{dt} + \begin{bmatrix} \chi_v \\ \chi_f \end{bmatrix}^T ((\mathbf{Q}_N - (\mathbf{Q}_N)^T \circ \mathbf{F}_S) \mathbf{1} + \chi_f \mathbf{W}_f \text{diag}(\hat{\mathbf{n}}) \mathbf{f}^* = 0$$

This was expanded to non-linear stable flux reconstruction³ (NSFR) schemes using a filtered DG form. Where $\mathbf{M}$ is replaced with $\mathbf{M} + \mathbf{K}$. $\mathbf{K}$ is defined as the following

$$\mathbf{K} = c(\mathbf{D}^p)^T \mathbf{M}(\mathbf{D}^p)$$

²Chan, Jesse. *Journal of Scientific Computing*. 2019.

³Cicchino, Alexander, Nadarajah, Siva, and Del Rey Fernández, David C. *Journal of Computational Physics*. 2022.

Entropy Stable Discontinuous Galerkin

There are a few conditions to ensure entropy stability:

- An entropy conservative flux must be used
- The flux must be calculated using the entropy projected conservative variables, $\tilde{\mathbf{u}} = \mathbf{u}(\tilde{\mathbf{v}})$
- The nodal entropy variables are calculated from the quadrature entropy variables using a projection operator, $\mathbf{\Pi}_V = \mathbf{M}^{-1} \chi_V^T \mathbf{W}$

→ Local Entropy Conservation: $\mathbf{1}^T \mathbf{W} \frac{dS(\mathbf{u}_q)}{dt} = \mathbf{1}^T \mathbf{W}_f \text{diag}(\hat{\mathbf{n}}) (\psi(\tilde{\mathbf{u}}_f) - \tilde{\mathbf{v}}_f^T \mathbf{f}^*)$

To achieve this, we pre-multiply by the nodal entropy variables

Entropy Stability in ROMs

- In a ROM, conservative quantities are conserved, but the entropy of the original problem is not.
- If we multiply our reduced-order equation by the entropy variables, no statement of conservation is found.

Entropy Stable Projection ROM

In order to preserve entropy in our model, we change the projection operator used to transform the L^2 projection operator to project it into the reduced order space⁴.

$$\Pi = \Pi_{ROM}\Pi_V = (\mathbf{V}^T \mathbf{M} \mathbf{V})^{-1} \mathbf{V}^T \chi_V \mathbf{W}$$

The individual components are shown below

- $\Pi_{ROM} = (\mathbf{V}^T \mathbf{M} \mathbf{V})^{-1} \mathbf{V}^T \mathbf{M}$
- $\Pi_V = \mathbf{M}^{-1} \chi_V^T \mathbf{W}$

⁴Chan, Jesse. *Journal of Computational Physics*. 2020.

Entropy Stable Projection ROM

This also achieves local conservation by pre-multiplying by the transpose of the entropy variables. The global statement of entropy conservation can then be written as:

$$\sum_{k=1}^K \mathbf{1}^T J^k \mathbf{W} \frac{dS(\mathbf{u}_q)}{dt} = \sum_{k=1}^K \mathbf{1}^T \mathbf{W}_f \text{diag}(\hat{\mathbf{n}}) (\psi(\tilde{\mathbf{u}}_f) - \tilde{\mathbf{v}}_f^T \mathbf{f}^*)$$

→ This requires that **V accurately** describes the entropy variables

- To accurately describe the entropy variables in our basis, we turn to a Proper Orthogonal Decomposition⁵ (POD) and method of snapshots for the basis.
- A Singular Value Decomposition is used on a collection of the nodal conservative variables and nodal entropy variables for selected time steps.

$$\text{svd}([\mathbf{u}_h^0 \quad \dots \quad \mathbf{u}_h^n \quad \mathbf{v}_h^0 \quad \dots \quad \mathbf{v}_h^n]) = \mathbf{V}\mathbf{\Sigma}\mathbf{Z}^T$$

⁵Holmes, Philip, Lumley, John L., and Berkooz, Gal. 1996.

Test Cases

- ① 1D Shock Tube
- ② 2D Riemann Problem
- ③ 2D Isentropic Advecting Vortex

→ All cases are using the Euler Equations

1D Shock

This 1D shock⁶ contains a wall boundary condition on either side of the domain

- c+ Flux Reconstruction⁷
- Lax-Friedrichs Dissipation
- Chandrashekar Flux with Ranocha fix
- 3 Stage Explicit SSPRK
- 2nd order Polynomial
- GL Solution Nodes
- Starting CFL = 0.9

⁶Chan, Jesse. *Journal of Computational Physics*. 2020.

⁷Castonguay, Patrice. 2012.

Figure: Density along Domain

1D Shock: Stability

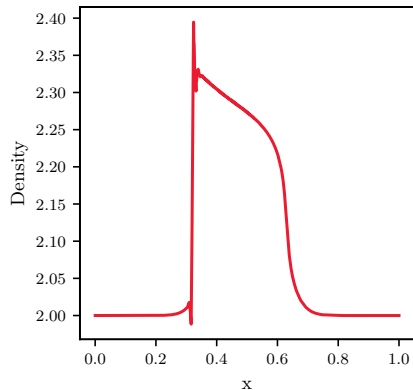


Figure: Density of p2, CFL 0.9, c+ 1D at $t = 0.75s$

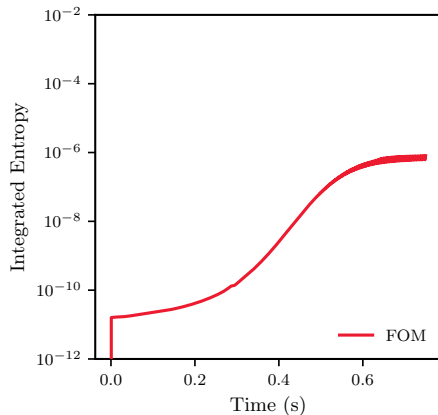


Figure: Integrated Entropy of p2, CFL 0.9, c+, 1D Shock

1D Shock: Stability

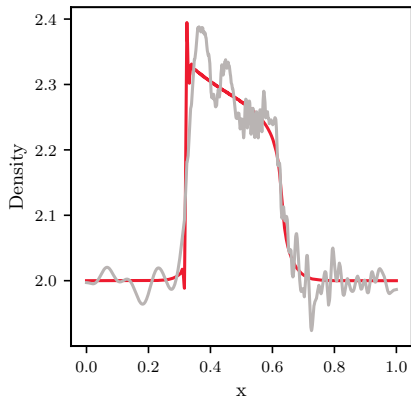


Figure: Density of p2, CFL 0.9, c+ 1D at $t = 0.75s$

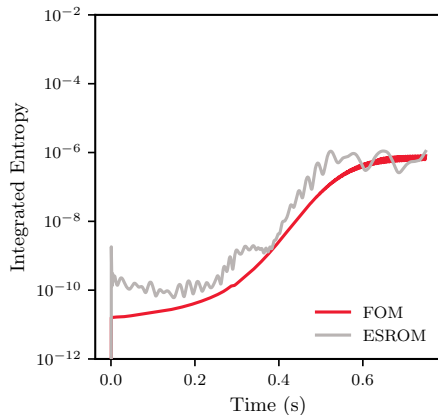


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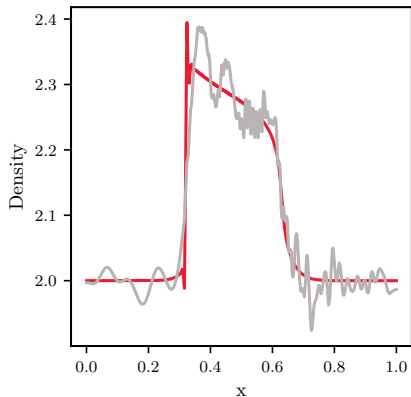


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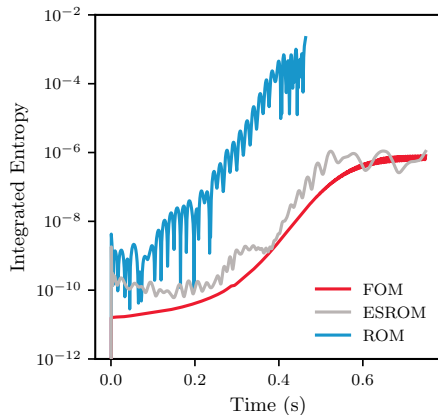


Figure: Integrated Entropy of p2, CFL 0.9, c+, 1D Shock

1D Shock: Singular Values

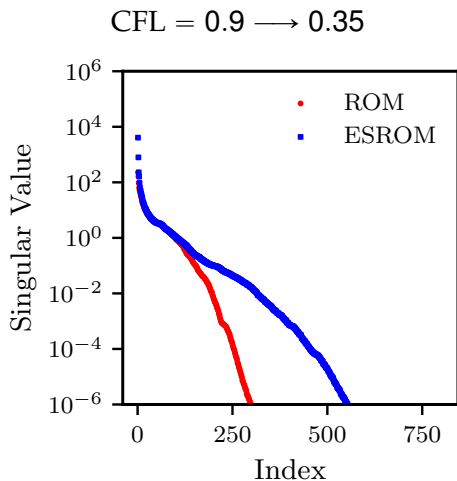


Figure: Singular Values, p2, CFL 0.35, c+, 1D Shock

1D Shock: Singular Values

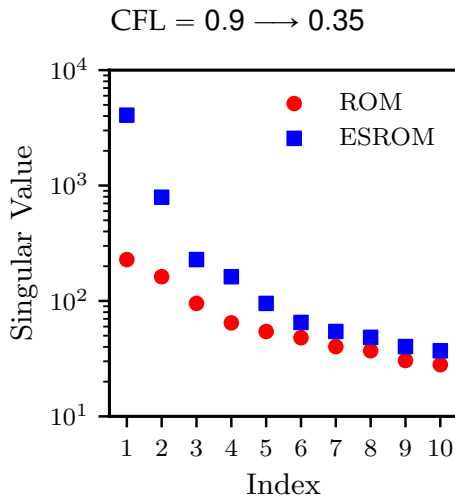


Figure: Singular Values, p2, CFL 0.35, c+, 1D Shock Zoomed in

1D Shock: Lowering CFL

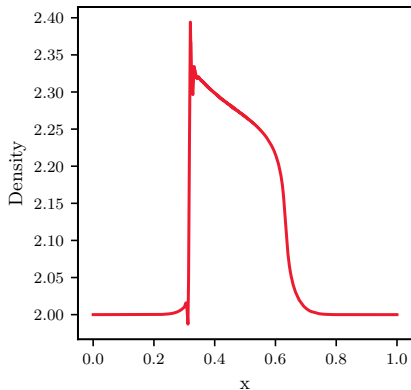


Figure: Density of p2, CFL 0.35, c+ 1D at $t = 0.75s$

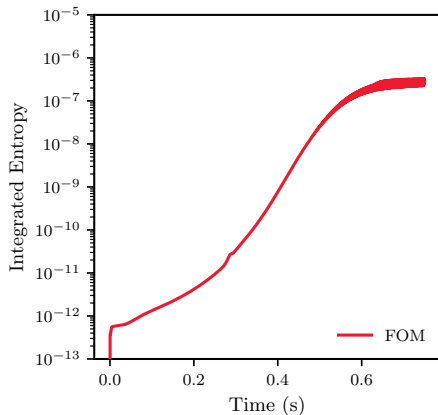


Figure: Integrated Entropy of p2, CFL 0.35, c+, 1D Shock

1D Shock: Lowering CFL

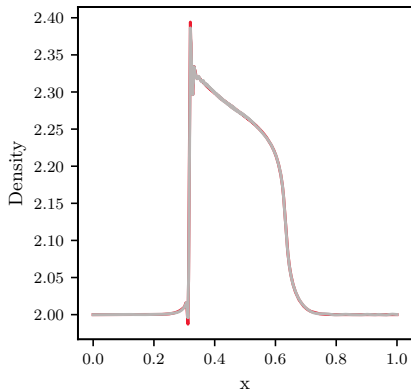


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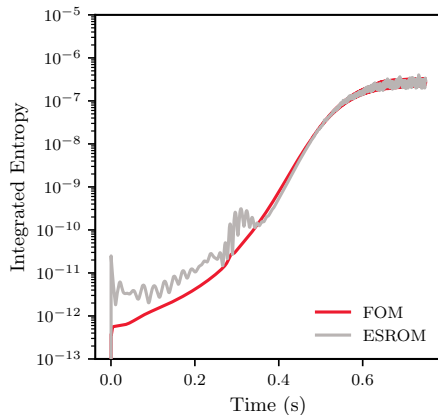


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1D Shock: Lowering CFL

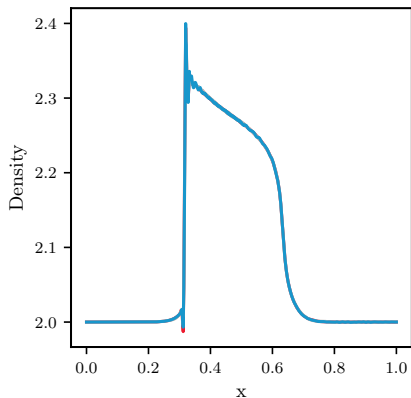


Figure: Density of p2, CFL 0.35, c+ 1D at $t = 0.75s$

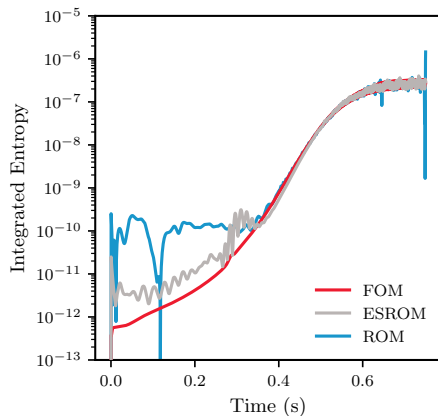


Figure: Integrated Entropy of p2, CFL 0.35, c+, 1D Shock

1D Shock: Increasing Polynomial Order

$$p = 2 \longrightarrow 6, c = c+ \longrightarrow c_{Hu}^8, CFL = 0.35 \longrightarrow 0.1$$

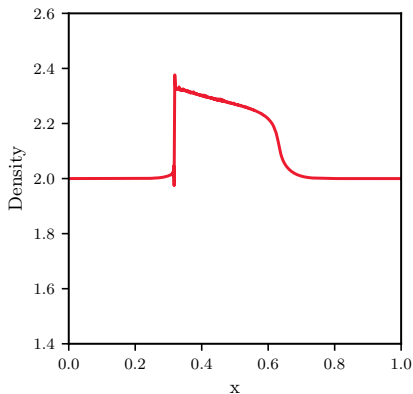


Figure: Density of p6, CFL 0.1, c_{Hu} 1D at $t = 0.75s$

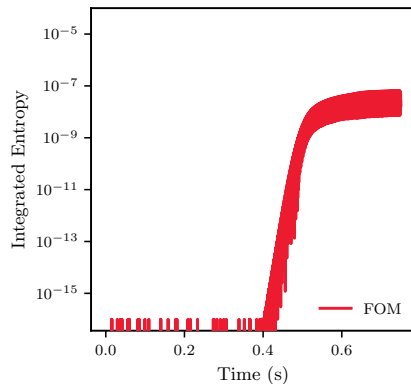


Figure: Integrated Entropy of p6, CFL 0.1, c_{Hu} , 1D Shock

⁸Huynh, Hung T. 2007.

1D Shock: Increasing Polynomial Order

$$p = 2 \longrightarrow 6, c = c+ \longrightarrow c_{Hu}^9, CFL = 0.35 \longrightarrow 0.1$$

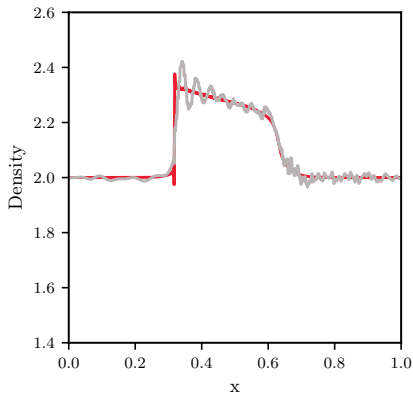


Figure: Density of p6, CFL 0.1, c_{Hu} 1D at $t = 0.75s$

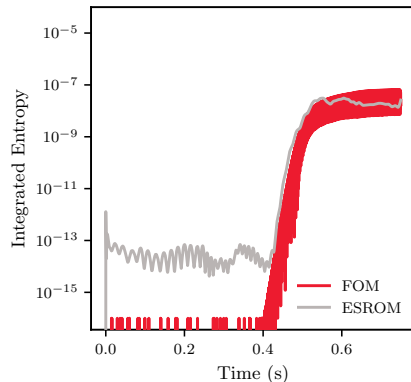


Figure: Integrated Entropy of p6, CFL 0.1, c_{Hu} , 1D Shock

⁹Huynh, Hung T. 2007.

1D Shock: Increasing Polynomial Order

$$p = 2 \longrightarrow 6, c = c+ \longrightarrow c_{Hu}^{10}, CFL = 0.35 \longrightarrow 0.1$$

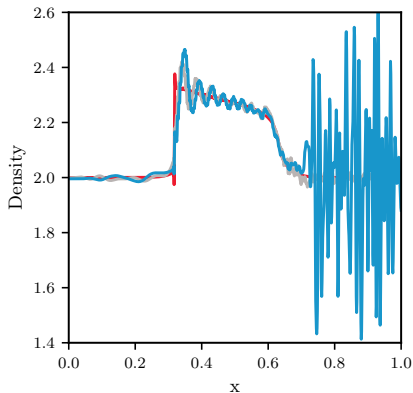


Figure: Density of p6, CFL 0.1, c_{Hu} 1D at $t = 0.75s$

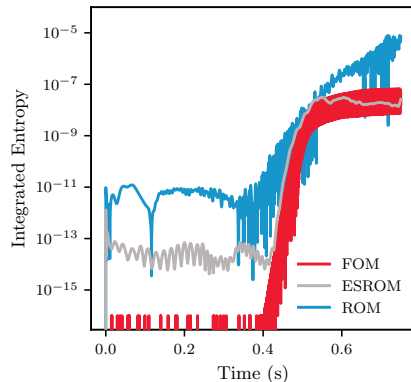


Figure: Integrated Entropy of p6, CFL 0.1, c_{Hu} , 1D Shock

¹⁰Huynh, Hung T. 2007.

2D Riemann Problem

The 2D Riemann problem¹¹ has an under-resolved shock that propagates as time advances.

- $c+$ Flux Reconstruction
- Lax-Friedrichs Dissipation
- Chandrashekar Flux with Ranocha fix
- 3 Stage Explicit SSPRK
- 3rd order Polynomial
- GL Solution Nodes
- CFL = 0.025



Figure: 2D Riemann Test Case at Final time, $t = 0.25s$

¹¹Chan, Jesse. *Journal of Computational Physics*. 2020.

2D Riemann Problem: Singular Values

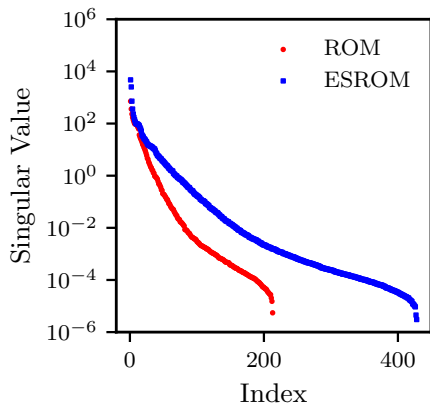


Figure: Singular Values of p_3 , CFL 0.025, c+, 2D Riemann

2D Riemann Problem: Singular Values

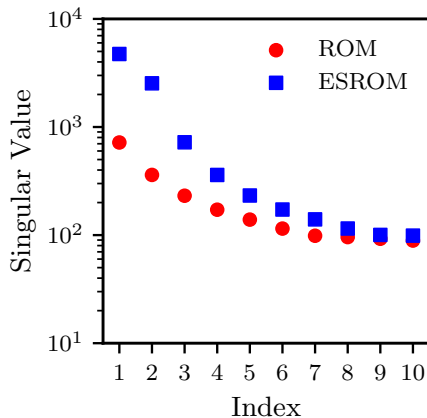


Figure: Singular Values of p3, CFL 0.025, c+, 2D Riemann Zoomed in

2D Riemann Problem: Integrated Entropy

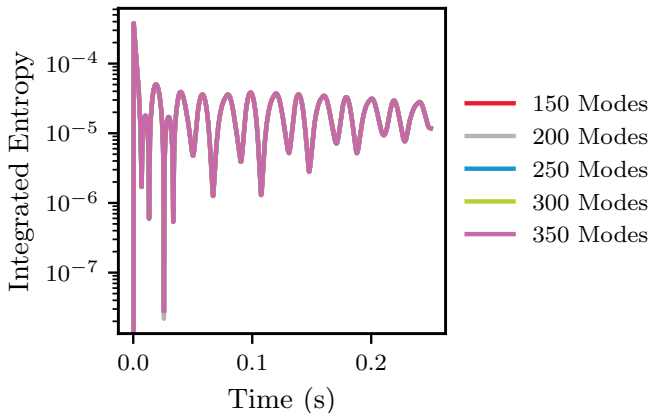


Figure: Integrated Entropy p3, CFL 0.025, c+, 2D Riemann

Isentropic Vortex

To test for the models performance on a variety of grid sizes, we test an isentropic vortex advection¹² across $x = y$

- Chandrashekar Flux with Ranocha fix
- GLL Solution Nodes

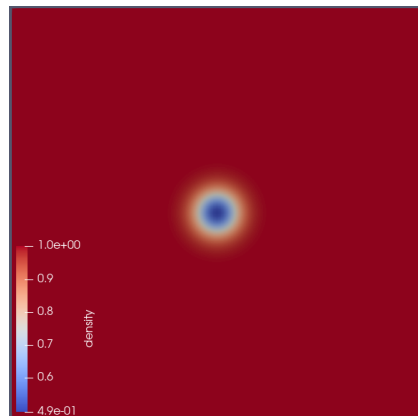


Figure: Isentropic vortex advection on a $[-10,10]^2$ domain.

¹²Wang, Zhijian J et al. *International Journal for Numerical Methods in Fluids*. 2013.

Isentropic Vortex: Grid Refinement

Convergence order of the fully-discrete scheme while refining an $n \times n$ grid:

Scheme	n	L^2 error	L^2 rate	ESROM L^2 error	ESROM L^2 rate
SSPRK3, $p = 2$	8	6.532E-1	-	6.531E-1	-
	16	3.885E-1	0.75	3.888E-1	0.75
	32	3.889E-2	3.32	3.890E-2	3.32
	64	6.309E-3	2.62	6.303E-3	2.63
	128	8.006E-4	2.98	8.106E-4	2.96
RK4, $p = 3$	8	4.009E-1	-	4.009E-1	-
	16	3.464E-2	3.53	3.432E-2	3.53
	32	3.217E-3	3.43	3.227E-3	3.43
	64	3.616E-4	3.15	3.731E-4	3.11

Isentropic Vortex: Integrated Entropy

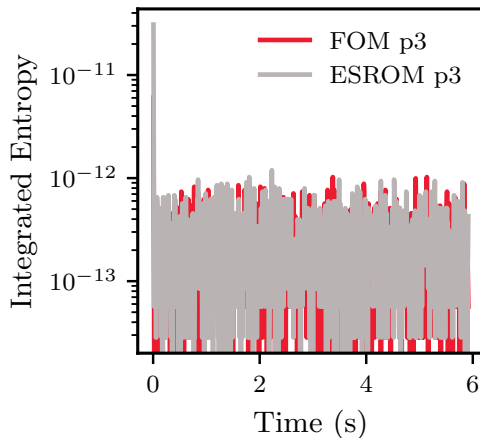


Figure: Integrated Entropy for p_3 Vortex Advection 64^2 cells

Conclusions and Future Work

- Entropy stable ROMs track the full order solution
- They remain entropy stable for the entire length of the simulation

Future Work

- We are interested in extending this to **hyper-reduction** to achieve cost savings.

Acknowledgements

We are grateful for our sources of funding, Bombardier and **McGill University**.

All test cases use the `PHILIP` library (Shi-Dong and Nadarajah, 2021). Some results use the Narval compute cluster, supported by **Calcul Québec** and the **Digital Research Alliance of Canada**.



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Alliance of Canada

Alliance de recherche
numérique du Canada

Questions?

Thank you for your attention!

Singular Vector: 1D Shock

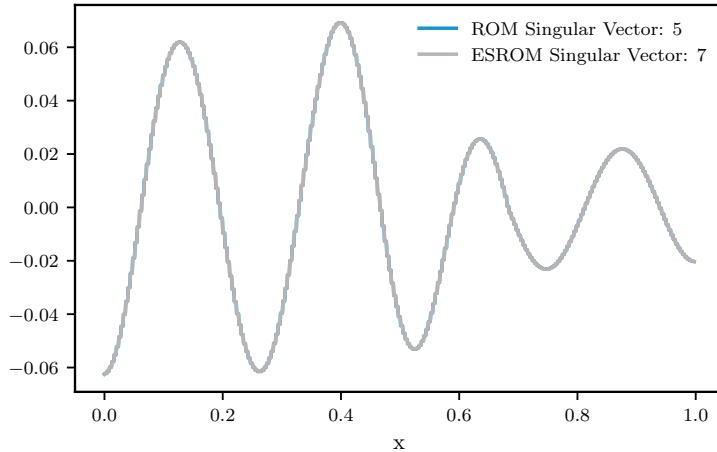


Figure: Fifth Singular Vector for Density of the ROM and ESROM PODs